

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA5117 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5 TOTAL MARKS: 50

Annexure A

Comparative Analysis

Code of Conduct:

I Daniel Jesuraji J (192565085) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Annexure A

Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.

Authentication Header (AH) and Encapsulating Security Payload (ESP) are the two main IPSec protocols used to protect IP communication. They differ mainly in the security services they provide, performance overhead, and suitability for different network scenarios.

Comparison of AH and ESP

Feature	Authentication Header (AH)	Encapsulating Security Payload (ESP)
Authentication	Yes	Yes
Data Integrity	Yes	Yes
Confidentiality	No	Yes
Encryption	Not provided	Provided
Anti-Replay Protection	Yes	Yes
Performance Overhead	Lower	Higher due to encryption

Feature	Authentication Header (AH)	Encapsulating Payload (ESP)	Security
Protection	Protects packet integrity and authentication	Protects data confidentiality, integrity, and authentication	
Suitable For	Traffic requiring authentication and integrity	Confidential and sensitive communication	

1. Security Services

AH provides:

- Data integrity
- Data-origin authentication
- Anti-replay protection

However, AH **does not encrypt the data**, so the contents of the communication remain visible.

ESP provides:

- Data confidentiality
- Data integrity
- Data-origin authentication
- Anti-replay protection

Therefore, ESP provides more comprehensive protection when confidential information is transmitted.

2. Performance Overhead

AH generally has **lower processing overhead** because it does not perform encryption.

ESP has **higher overhead** because encryption and decryption require additional processing. However, modern systems can efficiently perform encryption using

hardware acceleration.

Therefore:

AH → Lower overhead → Less processing

ESP → Higher overhead → More processing but stronger protection

3. Suitability for Different Network Scenarios

AH is suitable when:

- Data confidentiality is not required.
- Authentication and integrity are the primary requirements.
- The organization needs protection against modification of IP packets.

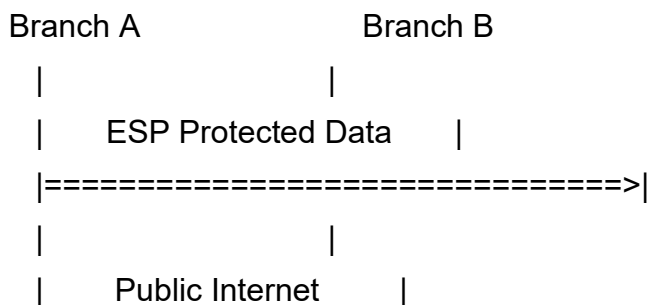
ESP is suitable when:

- Sensitive information is transmitted.
- Confidentiality is required.
- VPN communication is being established.
- Business, financial, or private data must be protected.

4. Suitability for VPN Deployment

For a VPN deployment, **ESP is more appropriate than AH** because VPN traffic commonly requires confidentiality in addition to integrity and authentication.

A typical VPN configuration is:



ESP can encrypt the transmitted data, preventing unauthorized users on the public Internet from reading the contents.

Conclusion

Although AH provides authentication, integrity, and anti-replay protection with lower overhead, it does not provide confidentiality. **ESP is more suitable for VPN deployment** because it provides encryption along with integrity, authentication, and anti-replay protection. Therefore, for modern enterprise VPNs carrying sensitive information, **ESP is generally the preferred IPSec protocol**.

2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.

IPSec provides two modes of operation: **Transport Mode** and **Tunnel Mode**. They differ in how much of the IP packet is protected and are suitable for different communication scenarios. Transport Mode is mainly used for **host-to-host communication**, while Tunnel Mode is mainly used for **network-to-network communication**.

Comparison of Transport Mode and Tunnel Mode

Feature	Transport Mode	Tunnel Mode
Primary Use	Host-to-host communication	Network-to-network communication
Protection	Protects the IP payload	Protects the complete original IP packet
Original IP Header	Remains visible	Encapsulated and protected
New IP Header	Not normally added	Added for routing

Feature	Transport Mode	Tunnel Mode
Security Level	Suitable for direct communication	Provides stronger network-level protection
Common Application	Host-to-host IPSec	Site-to-site VPN
Overhead	Lower	Higher
Suitable For	Direct host communication	Branch-office and VPN communication

1. Transport Mode

In **Transport Mode**, the original IP header remains intact while the payload is protected by IPSec.

Original IP Header

+

IPSec Header

+

Protected Payload

It is mainly used when **two individual hosts communicate directly**.

Advantages:

- Lower overhead.
- Efficient for direct host-to-host communication.
- Requires less processing than tunnel mode.

Limitation:

The original IP header is not completely protected, so some routing information remains visible.

2. Tunnel Mode

In **Tunnel Mode**, the entire original IP packet is encapsulated inside a new IP packet.

New IP Header

+

IPSec Header

+

Original IP Header

+

Original Payload

It is mainly used for **network-to-network communication**, particularly in site-to-site VPNs.

Advantages:

- Protects the complete original IP packet.
- Hides the internal source and destination addresses.
- Suitable for communication between branch-office networks.
- Provides strong protection over the public Internet.

Limitation:

Tunnel Mode introduces additional processing and packet overhead because a new IP header is added.

3. Host-to-Host vs Network-to-Network

Host-to-Host:

Host A

|

| Transport Mode

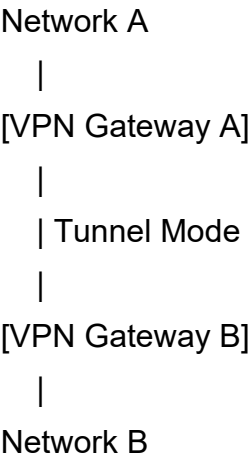
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Host B

Transport Mode is effective because the communication occurs directly between

two hosts.

Network-to-Network:



Tunnel Mode is more effective because it protects the original packets traveling between two networks.

4. Effectiveness Comparison

Communication	Recommended Mode	Reason
Host-to-host	Transport Mode	Efficient direct host communication
Server-to-server	Transport Mode	Lower overhead
Network-to-network	Tunnel Mode	Protects complete original packets
Site-to-site VPN	Tunnel Mode	Provides secure network-level communication
Remote network	Tunnel Mode	Protects traffic across public networks

Communication	Recommended Mode	Reason
access		

Conclusion

Transport Mode is more suitable for **host-to-host communication** because it provides protection with lower overhead. **Tunnel Mode** is more suitable for **network-to-network communication and VPNs** because it encapsulates and protects the complete original IP packet. Therefore, for a corporate site-to-site VPN, **Tunnel Mode is the preferred choice**, while Transport Mode is appropriate for direct host-to-host IPSec communication.

3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system

Pretty Good Privacy (PGP) and **S/MIME (Secure/Multipurpose Internet Mail Extensions)** are technologies used to secure email communication. Both support encryption and digital signatures, but they differ in **key management, trust models, scalability, and enterprise usability**.

Comparison of PGP and S/MIME

Feature	PGP	S/MIME
Key Management	User-managed public and private keys	Certificate-based key management

Feature	PGP	S/MIME
Trust Model	Web of Trust	Certificate Authority (CA) based
Authentication	Public-key based	Digital certificates
Digital Signature	Supported	Supported
Encryption	Supported	Supported
Scalability	Moderate	High
Enterprise Usability	Requires more user involvement	Easier for centralized administration
Centralized Management	Limited	Strong
Suitable Environment	Individual users and decentralized environments	Corporate and enterprise environments

1. Key Management

PGP:

- Users generate and manage their own key pairs.
- Public keys are exchanged between users.
- Users are responsible for verifying and maintaining keys.
- Key management can become difficult when the number of users increases.

S/MIME:

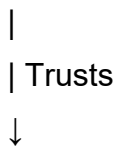
- Uses digital certificates containing public keys.
- Certificates are normally issued by a trusted **Certificate Authority (CA)**.
- Organizations can centrally manage certificates.
- Certificate renewal and revocation can be managed systematically.

Therefore, **S/MIME provides easier centralized key management for large organizations.**

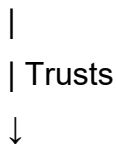
2. Trust Models

PGP generally uses a **Web of Trust** model.

User A



User B

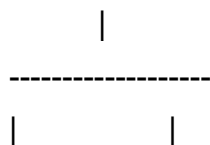


User C

Trust is established through users verifying and signing each other's public keys.

S/MIME uses a **Certificate Authority-based trust model.**

Certificate Authority



User A Certificate User B Certificate

The CA establishes trust by issuing and validating digital certificates.

3. Scalability

PGP can become difficult to manage in large organizations because users are responsible for managing keys and establishing trust.

S/MIME is more scalable because certificates can be centrally issued, managed, renewed, and revoked.

Therefore:

PGP → Moderate scalability

S/MIME → High scalability

4. Enterprise Usability

For a corporate environment, S/MIME generally provides better usability because it can be integrated with organizational email infrastructure and centralized certificate-management systems.

PGP provides flexibility and user control but may require more user involvement in key management and trust verification.

5. Recommendation for Corporate Email

For a **corporate email system**, **S/MIME is the recommended approach** because:

1. It provides centralized certificate management.
2. It uses a structured CA-based trust model.
3. It is more scalable for large organizations.
4. It supports encryption and digital signatures.
5. It is suitable for enterprise email integration.
6. Certificate renewal and revocation can be managed systematically.

Conclusion

Both PGP and S/MIME provide secure email communication through encryption and digital signatures. However, **S/MIME is more suitable for a corporate email system** because of its certificate-based trust model, centralized key management, scalability, and enterprise usability. PGP is more suitable when decentralized trust and greater user control over keys are preferred.

4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.

Rule-based and machine learning-based techniques are commonly used to detect spam emails. Rule-based systems depend on predefined conditions, while machine learning-based systems learn patterns from email data. The assessment specifically requires comparison of their effectiveness in handling **evolving spam patterns** and **minimizing false positives**.

Comparison of Rule-Based and Machine Learning-Based Techniques

Feature	Rule-Based Detection	Machine Learning-Based Detection
Detection Method	Uses predefined rules	Learns patterns from training data
Adaptability	Low	High
New Spam Patterns	Difficult to detect	Better at detecting new patterns
False Positives	Can be higher	Can be reduced with proper training
Maintenance	Rules must be updated manually	Model can be retrained
Complexity	Low	Higher

Feature	Rule-Based Detection	Machine Learning-Based Detection
Processing	Generally simple and fast	Requires feature processing and model prediction
Scalability	Limited for changing threats	Better for large and changing datasets

1. Rule-Based Spam Detection

Rule-based systems identify spam using predefined conditions.

Examples of rules include:

- Suspicious keywords.
- Known spam sender addresses.
- Suspicious email patterns.
- Unwanted links or phrases.
- Repeated spam characteristics.

Advantages:

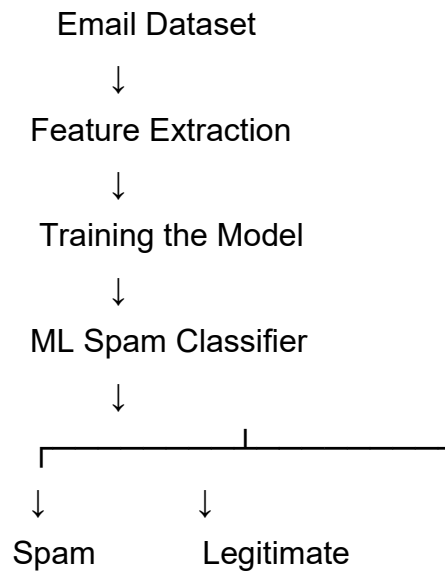
1. Simple to implement.
2. Easy to understand.
3. Fast detection.
4. Effective against known spam patterns.

Limitations:

1. Requires frequent rule updates.
2. May fail against new spam techniques.
3. Attackers can modify messages to bypass rules.
4. Strict rules may incorrectly classify legitimate emails as spam.

2. Machine Learning-Based Spam Detection

Machine learning systems learn patterns from previously classified emails and use those patterns to classify new messages.



Possible features include:

- Email text.
- Keywords.
- Sender information.
- URL characteristics.
- Message structure.
- Header information.

Advantages:

1. Can learn complex spam patterns.
2. Adapts better to changing spam techniques.
3. Can process large volumes of emails.
4. Can reduce false positives when properly trained.

Limitations:

1. Requires suitable training data.
2. Model performance depends on data quality.
3. Requires periodic retraining.
4. More complex than rule-based detection.

3. Handling Evolving Spam Patterns

Rule-based systems are less effective against evolving spam because attackers can change words, sender addresses, and message structures to bypass predefined rules.

Machine learning systems are generally more adaptable because they can learn new patterns from updated datasets.

Rule-Based

Known Pattern → Rule → Detection



New Pattern → May Bypass Rule

Machine Learning

Training Data → Model → Pattern Learning



New Pattern Detection

Therefore, **machine learning is more suitable for continuously changing spam patterns.**

4. Minimizing False Positives

A false positive occurs when a legitimate email is incorrectly classified as spam.

Rule-Based Approach:

- Use carefully designed rules.
- Avoid overly broad keywords.
- Maintain trusted sender lists.
- Regularly review and update rules.

Machine Learning Approach:

- Use a diverse and accurately labelled dataset.
- Select relevant features.
- Regularly retrain the model.
- Evaluate precision and false-positive rate.
- Allow feedback on incorrectly classified emails.

Conclusion

Rule-based detection is **simple, fast, and effective against known spam**, but it requires continuous manual updates and may struggle with evolving attacks. Machine learning-based detection is **more adaptable to changing spam patterns** and can minimize false positives when trained and maintained properly. Therefore, for a modern email system, **machine learning-based detection is generally more suitable**, while rule-based techniques can be used as an additional layer of protection.

- 5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

Key management is an important part of IPSec because encryption and authentication depend on securely established keys. **Manual key management** requires administrators to configure keys and security parameters manually, whereas **automated key management**, such as using **IKE (Internet Key Exchange)**, automatically negotiates and manages the required security associations and keys.

Comparison of Manual and Automated Key Management

Feature	Manual Key Management	Automated Key Management (IKE)
Key Configuration	Manually configured	Automatically negotiated
Key Renewal	Manual	Automatic
Security	More prone to human errors	More secure with automated procedures

Feature	Manual Management	Key	Automated Management (IKE)	Key
Scalability	Low		High	
Operational Complexity	High		Lower	
Administration	Requires continuous manual effort		Requires less manual intervention	
Key Synchronization	Difficult		Automatically coordinated	
Suitability	Small and static networks		Large enterprise networks	

1. Manual Key Management

In manual key management, network administrators configure encryption keys and security parameters on IPSec devices.

Advantages:

1. Simple for very small networks.
2. Does not require an automated key-exchange mechanism.
3. Provides direct administrator control.

Limitations:

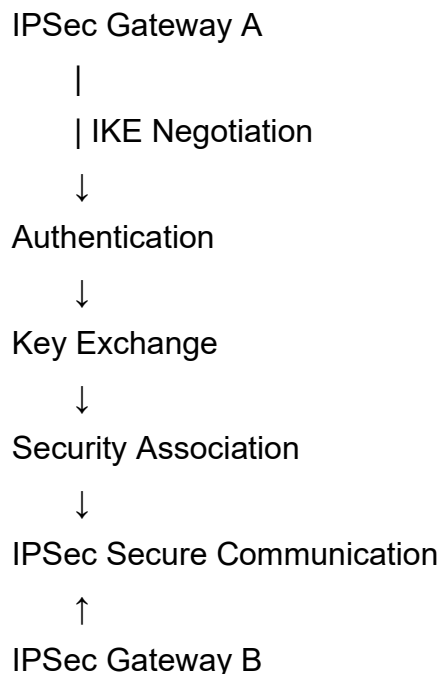
1. Keys must be configured manually.
2. Key renewal requires manual intervention.
3. Human errors can result in mismatched keys.
4. Managing many devices becomes difficult.

5. Compromised keys may remain active until manually replaced.

Therefore, manual key management becomes impractical as the organization grows.

2. Automated Key Management Using IKE

IKE (Internet Key Exchange) automates the negotiation and establishment of security parameters and keys for IPSec communication.



The process reduces manual configuration and allows keys to be renewed automatically.

Advantages:

1. Automatic key exchange.
2. Automatic key renewal.
3. Reduced configuration errors.
4. Better scalability.
5. Reduced administrative effort.
6. More suitable for large organizations.

3. Impact on Security

Manual Management:

- Higher risk of human error.
- Manual key replacement can be delayed.
- Key mismatches may interrupt secure communication.

Automated Management:

- Reduces human configuration errors.
- Supports automatic key renewal.
- Provides coordinated key establishment.
- Improves the overall security of large IPSec deployments.

Therefore, automated key management provides better security for large and dynamic environments.

4. Impact on Scalability

Manual management becomes increasingly difficult when the number of IPSec gateways increases.

Small Network

Few Devices → Manual Management → Possible

Large Network

Many Devices → Manual Management → Difficult

Large Network

Many Devices → IKE Automation → Scalable

IKE can automatically establish and renew keys for multiple IPSec connections, making it more suitable for large organizations.

5. Operational Complexity

Manual key management requires administrators to configure, monitor, renew, and replace keys. This increases operational workload.

Automated key management reduces these activities by automatically negotiating and renewing keys. Administrators mainly need to configure policies, authentication,

and monitoring mechanisms.

Conclusion

Manual key management may be suitable for **small and static networks**, but it becomes difficult to maintain as an organization grows. **Automated key management using IKE is more suitable for large organizations** because it improves security, scalability, and operational efficiency while reducing manual configuration and key-management errors. Therefore, enterprises should prefer **automated IPSec key management** for large-scale deployments.